Lecture 1: Data Structure and Baseline Forecasts

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A forecast is a statement about an outcome made before that outcome is observed. A model is a rule that connects available information to an outcome. Session 0 introduced the purpose and structure of the course. Here we specify a forecasting question, arrange the observations, calculate simple predictions, and check their errors.

1 Write Down the Forecasting Question

A variable is a quantity that can take different values, such as daily sales. An observation is one recorded value, such as 120 sandwiches sold on Monday. A target variable is the quantity you want to predict. The forecast origin is the moment when you make the forecast. The forecast horizon is how far ahead you are looking.

Imagine it is Monday at six in the evening. You predict Tuesday's sandwich sales.

Part of the question	Shop forecast
Target	Number of sandwiches sold on Tuesday
Origin	Monday, 6 p.m.
Horizon	One day ahead
Available information	Sales through Monday and promotions announced by Monday evening
Evaluation	Compare the prediction with Tuesday's recorded sales
Decision	How much bread to order

The unit of measurement tells you what a number counts. Here it is sandwiches, not dollars or percentage growth. Predicting 120 sandwiches is a different task from predicting $120 of revenue.

Evaluation means checking a forecast against the outcome after the outcome becomes available. To evaluate this forecast, save Monday's prediction before Tuesday arrives. Replacing it on Tuesday evening would erase the prediction you actually made.

1.1 Read the Meaning Before the Symbols

Let t label the current period. A period can be a day, month, or quarter. The observed value is written as

$$y_t.$$

Here y names the quantity and t tells us when it was observed. If Monday is period t, then $y_t = 120$ means that Monday's recorded sales were 120 sandwiches.

Fitting a model means choosing its adjustable numbers using recorded data. A hat above a value indicates a model-produced value. Two uses need to be kept separate:

Symbol	Meaning
y_t	The actual recorded outcome for period t.
$\hat{y}_t$	A fitted value: the model's estimate of the outcome for a period already represented in the data being fitted.
$\hat{y}_{t+1\|t}$	A forecast made at time t for the next period, $t+1$.

A fitted value describes what a model produces for an observation in the data it was built to fit. A forecast value is produced before the future outcome is observed. Both can wear a hat. The timing tells you which task the model is doing.

In $\hat{y}_{t+1|t}$, the part before the vertical bar identifies the period being predicted. The part after the bar identifies when the prediction is made. If your forecast is 125 and Tuesday's recorded sales turn out to be 130, the forecast remains 125 and the observed outcome is 130.

For a longer horizon, we write $\hat{y}_{t+h|t}$. The letter h counts periods ahead. A two-day forecast made on Monday concerns Wednesday. It must still use information available on Monday.

2 What Information Can You Use?

An information set is the collection of facts available at the forecast origin. We write the information set at time t as $\mathcal{I}_t$. It is like a folder you close when the forecast is made. Information arriving afterward belongs to a later folder.

For the Monday-evening forecast:

Information	Available by Monday evening?
Monday's recorded sales	Yes
Tuesday's promotion, announced last Friday	Yes
A Tuesday weather forecast issued Monday morning	Yes
Tuesday's realized rainfall	No
Tuesday's recorded sales	No
A corrected sales file released on Wednesday	No, if that correction was unavailable on Monday

A future event can be usable if its schedule is already known. You know that Tuesday is a holiday before Tuesday begins. You do not yet know Tuesday's realized sales. Always ask when the particular piece of information became available.

Data leakage occurs when information that should be unavailable to a model enters its construction or evaluation. Using Tuesday's realized sales to improve a prediction supposedly made Monday is leakage. It can make a recorded test look excellent while giving the manager no usable Monday forecast. Lecture 2 examines this problem in detail.

3 Arrange the Data Before Choosing a Model

Data are recorded observations. An entity is the person, business, product, or other unit being observed. The arrangement of entities and dates is the data structure.

3.1 Cross-Sectional Data: Many Entities at One Time

Cross-sectional data record different entities at the same time or over the same reference period.

Imagine you collect Monday's sales from three shops:

Shop	Monday sales
A	120
B	90
C	150

You can compare shops on Monday. This table alone cannot tell you whether Shop A's sales have been rising over the last month because it contains only one date for that shop.

3.2 Time-Series Data: One Entity Across Dates

Time-series data record a quantity in time order for one entity or a defined aggregate. An aggregate combines several items into one measure, such as total sales across all shops.

For Shop A, the last three daily observations might be 100, 110, and 130. Their order matters. Reading them backward would turn an increase into a decrease. Frequency tells us how often observations are recorded: daily, weekly, monthly, or quarterly. A quarter is three months. A one-period horizon means one day for daily data and one month for monthly data.

3.3 Panel Data: Many Entities Across Dates

Panel data follow multiple entities over time.

Shop	Saturday	Sunday	Monday
A	100	110	130
B	80	85	90
C	140	135	150

Each row follows one shop across dates. Each column compares shops on one date. This single table contains both types of comparison. In finance, one stock's daily prices form a time series; daily prices for many identified stocks form a panel. A collection of stocks observed on one day is a cross section.

Check. Monthly sales for one hospital cafeteria are a time series. Monthly sales for ten identified hospital cafeterias form a panel. Ten cafeterias' sales for the same month form a cross section.

4 Build Simple Forecasts You Can Calculate by Hand

A model takes information available now and applies a rule to produce a prediction. A baseline gives you a simple reference: does a more complicated rule actually predict better?

Keep the same sales history: Saturday 100, Sunday 110, Monday 130. We are forecasting Tuesday on Monday evening. These numbers are illustrative, not measured results from a real shop.

4.1 The Latest-Value Forecast

The naive forecast, also called the latest-value forecast, uses the most recent observation as the next prediction:

$$\hat{y}_{t+1|t} = y_t.$$

The forecast for the next day equals the value observed today. Monday's sales were 130, so Tuesday's forecast is 130 sandwiches. This rule can be useful when the current level carries into the next period.

Persistence means that high or low values tend to carry forward through time. If a busy shop tends to remain busy the next day, the latest observation may contain useful information. Strong persistence does not mean that tomorrow will equal today exactly.

4.2 The Historical-Mean Forecast

The mean, or average, is the sum of the values divided by the number of values. The historical-mean forecast predicts the next outcome using the mean of observations available at the origin.

$$\bar{y}_t = \frac{y_1 + \cdots + y_t}{t}, \qquad \hat{y}_{t+1|t} = \bar{y}_t.$$

The bar marks an average. The subscript says that only observations available through the current period enter it. With our three observations,

$$\frac{100 + 110 + 130}{3} = 113.33.$$

This rule predicts about 113 sandwiches. Keep the decimal while comparing forecast methods; a physical order can later be rounded to whole portions. Averaging can reduce the effect of an unusually busy day, but it can also respond slowly when sales are genuinely rising.

4.3 A Moving Average

A moving average averages a fixed number of the most recent observations. That number is the window length. As a new observation arrives, the oldest one leaves the window.

Imagine four days with sales of 100, 110, 130, and 140. A three-day average first uses 100, 110, and 130, giving 113.33. When 140 arrives, the window uses 110, 130, and 140, giving 126.67. The value 100 drops out.

$$\hat{y}_{t+1|t} = \frac{y_t + y_{t-1} + y_{t-2}}{3}.$$

The right side contains the latest three observed values. Dividing their total by three produces the next-period forecast. The window moves one period when the origin moves one period.

Smoothing means reducing short-lived up-and-down movement in a series. Averaging usually produces a less jagged line than the raw observations. A longer window smooths more, but an old low value can keep the average below a newly higher sales level. Smoothing and forecasting are related uses of averaging: the average summarizes the observed window, and we can choose to use that summary as a forecast for the next period.

4.4 An Expanding Average

An expanding window keeps earlier observations and adds new ones. Using the same four values, the average expands from three observations to four:

$$\frac{100 + 110 + 130 + 140}{4} = 120.$$

The three-day moving average is 126.67, while the expanding average is 120. Both use only available data. They respond differently because one discards the oldest value and the other keeps it. Lecture 2 studies how to evaluate such choices over repeated forecast dates.

5 Measure How Far a Forecast Misses

A forecast error is the observed outcome minus the forecast. A positive error means the forecast was too low. A negative error means it was too high.

$$e_{t+1} = y_{t+1} - \hat{y}_{t+1|t}.$$

The first quantity on the right is what happened. The second is what we had predicted. Their difference is the miss. If Tuesday sales are 125, the latest-value forecast of 130 has error $125 - 130 = -5$. It overpredicted by five sandwiches. The historical-mean forecast of 113.33 underpredicted by about 11.67.

One day is too little evidence to establish which model is usually better. We need a sequence of forecasts made before their outcomes arrive. Suppose three saved forecasts give:

Day	Forecast	Actual sales	Error
1	100	110	10
2	120	115	−5
3	110	125	15

The average error is $(10 - 5 + 15)/3 = 6.67$. Errors of opposite signs can cancel. To describe the typical size of a miss, we need a measure that treats both directions as errors.

5.1 Mean Absolute Error

An absolute value is distance from zero: the absolute values of both -5 and 5 are 5. The mean absolute error, shortened to MAE, is the average size of the forecast misses without their signs.

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |e_i|.$$

Here N is the number of evaluated forecasts and i labels one of them. The summation sign means add the terms. The vertical bars remove the sign of each error; the final division produces their average. In our table,

$$MAE = \frac{10 + 5 + 15}{3} = 10 \text{ sandwiches.}$$

An MAE of 10 means that the average absolute miss was 10 sandwiches over these three days.

5.2 Root Mean Squared Error

Squaring a number means multiplying it by itself. It removes the sign and gives large misses extra weight. The root mean squared error, or RMSE, is the square root of the average squared errors. A square root reverses squaring; the square root of 100 is 10.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} e_i^2}.$$

Start with the errors, square them, take their average, and then take the square root:

$$RMSE = \sqrt{\frac{10^2 + (-5)^2 + 15^2}{3}} = \sqrt{116.67} \approx 10.80 \text{ sandwiches.}$$

A miss of 20 contributes 400 before averaging; a miss of 10 contributes 100. Doubling the miss makes its squared contribution four times as large. The final square root restores the target's original unit.

The choice of measure should reflect the decision. RMSE pays extra attention to large misses. MAE gives each additional unit of error the same weight. Neither measure by itself distinguishes the cost of leftover sandwiches from the cost of lost sales.

6 Give a Model a Fair Test

Estimation, also called fitting, means choosing a model's unknown numbers using data. The training sample contains observations used for that fitting. The test sample contains later observations reserved for checking forecasts. A sample is the set of observations used for a particular calculation.

In-sample performance describes performance on the data used to fit a model. Out-of-sample performance describes performance on observations excluded from that fitting. A forecasting test must also respect the forecast origin: all inputs must have been available when each prediction was made.

Imagine you have January through June sales. Use January through April to choose a forecasting rule. Save a forecast for May before seeing May sales. When May sales become available, record its error. A June forecast may use May sales, provided that update was part of the procedure you intended to evaluate. June's actual sales stay unavailable until the June forecast has been made.

A chronological holdout is a later period kept aside for evaluation. If you repeatedly use that period's scores to change the model, it is helping you choose the model. It then serves as validation rather than an untouched final test. Lecture 2 develops this distinction.

6.1 Compare Like with Like

To compare two models fairly, use the same target, unit, horizon, forecast dates, and allowed information. Imagine Model A predicts Shop A's daily sales with an MAE of 10 sandwiches. Model B predicts a hospital's monthly revenue with an MAE of $5,000. Those numbers cannot rank the models because they measure different tasks.

Even the same unit is insufficient. Errors of one percentage point for one stock's return and another stock's return concern different targets with potentially different difficulty. Compare models on the same forecasting question before attributing a score difference to the method.

A predictor observed in the same period as the target may explain co-movement without being usable for an earlier forecast. Today's realized stock return cannot be used in a forecast made yesterday. Moving the outcome into a test file does not repair that timing error.

6.2 Improvement Relative to a Baseline

A benchmark is a reference used for comparison. Here the benchmark is a baseline forecast that could actually have been made at each origin. A useful relative score compares the model's total squared forecast errors with the baseline's total squared forecast errors:

$$R^2_{\text{OS}} = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i^{\text{model}})^2}{\sum_{i=1}^{N}(y_i - \hat{y}_i^{\text{baseline}})^2}.$$

The subscript OS means out of sample. In this formula, the hatted values are saved forecasts for the same test observations; their origin labels are omitted to keep the comparison readable. The top total measures the model's misses. The bottom total measures the baseline's misses. Both must use the same target and dates, and the baseline's error total must be positive.

If the model's squared errors total 80 and the baseline's total 100, the score is $1 - 80/100 = 0.20$. The model reduced total squared error by 20 percent. A total of 120 gives $1-120/100 = -0.20$: the model was worse than the baseline. A negative score is possible and informative.

The familiar in-sample regression R^2 often uses deviations from the sample mean in its denominator. A real-time forecast comparison must state its actual baseline and how that baseline was calculated. A mean computed using the entire test period would be unavailable at the start of that period. Do not silently treat it as a forecast the manager could have made.

7 A Model That Learns a Relationship: Regression

A predictor, also called an input, is a variable supplied to a model to help predict the target. Regression describes a target as a systematic part related to predictors plus an unexplained part. A coefficient is a number governing that relationship; estimating a coefficient means choosing it from data.

Imagine the shop records tomorrow's planned lunch orders before closing today. Past records suggest that each additional planned order is associated with higher next-day sales. A one-input forecasting model can be written as

$$y_{t+1} = a + bx_t + u_{t+1}.$$

The left side is tomorrow's actual sales. The input x_t is the number of planned orders known today. The intercept a is the model's systematic value when the input is zero. The slope b is how much the systematic prediction changes when the input rises by one unit. The error term u_{t+1} represents what the systematic part does not explain about tomorrow's outcome.

Suppose the estimated intercept is 80 and the estimated slope is 2. If 20 planned orders are known today, the forecast is

$$\hat{y}_{t+1|t} = 80 + 2 \times 20 = 120.$$

The coefficient 2 means that one additional planned order is associated with two additional sandwiches in the fitted prediction. The model may capture group orders or other related activity. That association alone does not prove that changing the order count would cause exactly that increase in total sales.

A residual is the observed outcome minus the fitted value for an observation used in fitting. The model's underlying error term is unobserved; a calculated residual is an estimate of an unexplained part. A future forecast error compares a saved forecast with an outcome observed later. These differences have similar arithmetic, but their roles and timing differ.

7.1 Using the Series' Own Past: Autoregression

A lag is an earlier value of a variable. A one-period lag of today's sales is yesterday's sales. Autoregression is regression that uses the target's own past values as inputs. The prefix auto refers to the series itself.

An AR(1) model uses one lag:

$$y_{t+1} = c + \phi y_t + u_{t+1}.$$

Tomorrow's actual outcome is on the left. The current observed outcome is the input on the right. The coefficient ϕ describes how strongly the current value carries into the next period after allowing for the intercept c.

With estimated values $c = 20$ and $\phi = 0.8$, today's sales of 130 give a forecast of $20 + 0.8 \times 130 = 124$. The future error term is unknown, so we do not insert tomorrow's realized error into today's forecast. This calculation assumes the future error has mean zero given the information used by the model.

An AR(2) model uses two lags. An AR(p) model uses p lags, where p is the number of earlier periods included. More lags add inputs and coefficients to estimate. Whether that improves future forecasts needs a fair test.

8 What a Forecast Can and Cannot Tell the Manager

Uncertainty means that more than one future outcome remains possible given what is known. A probability is a number between zero and one describing the chance of an outcome. A distribution describes the possible outcomes and how probability is spread across them. A point forecast is one predicted number. An interval forecast gives a range together with a stated method or probability interpretation. The interval should be checked on new outcomes; a wide or narrow band drawn on a chart does not establish its reliability.

Imagine the shop's point forecast is 120. Two possible sales outcomes are 100 and 140. Ordering for 120 leaves 20 portions in one case and falls short by 20 in the other. If a shortage is more expensive than leftovers, the manager may choose an order above the point forecast. A forecast supplies information; the costs determine how to act on it.

The expected value is a probability-weighted average of possible outcomes. If sales are 100

or 140, each with probability one half, the expected value is $0.5 \times 100 + 0.5 \times 140 = 120$. Neither outcome needs to equal the average.

A conditional expectation is that average given specified information. If we know there is a promotion tomorrow, we should average over the outcomes possible with that information, rather than mixing promotion days with all other days. Its notation is

$$E[y_{t+1} \mid \mathcal{I}_t].$$

The expression inside identifies tomorrow's outcome. The information after the bar states what is known today. Under squared-error loss, meaning that a miss is scored by its square, this conditional average is the forecast with the smallest expected squared error when the distribution of possible outcomes is known. In practice we estimate relationships from limited data, so our fitted model can differ from that ideal average.

For the equally likely outcomes 100 and 140, predicting 120 gives squared errors of 400 and 400, averaging 400. Predicting 110 gives 100 and 900, averaging 500. This shows why the average is attractive under this scoring rule. It does not establish that 120 is the best inventory order when shortage and leftover costs differ.

9 A Complete First Forecasting Task

Imagine a cafeteria must plan tomorrow's lunch supply. Its last three observed daily sales are 90, 110, and 130. Tomorrow's promotion is already announced. Tomorrow's realized foot traffic is unknown.

1. Name the target, unit, origin, and horizon. The target is tomorrow's meals sold, measured in meals. The origin is today's planning time, and the horizon is one day.

2. List usable inputs. Past sales and the announced promotion are available. Tomorrow's realized foot traffic is unavailable.

3. Calculate two baselines. The latest-value forecast is 130. The mean of the three observations is $(90 + 110 + 130)/3 = 110$.

4. Save both forecasts. If tomorrow's sales are 120, the forecast errors are -10 and 10. Both forecasts miss by 10 meals on this day.

5. Repeat across later days using only information available at each origin. Compare the forecasts over the same dates and consider the cost of large mistakes.

Try it before reading the answers. A shop records 80, 100, and 120 sales. You forecast the next day after observing 120. What are the latest-value and three-day-mean forecasts? If the outcome is 110, which forecast has the smaller absolute error? Would a spreadsheet corrected two days later have been available at the origin?

Answers. The forecasts are 120 and 100. Their errors are -10 and 10, so their absolute

errors tie at 10. The later correction was unavailable unless the same corrected information was already known at the forecast origin.

10 Additional Concepts from the Original Lecture

The following concepts extend the same forecasting workflow. They are references for the wider Lecture 1 materials; the first forecasting task above does not require fitting these model families.

10.1 Levels, Changes, and Returns

A level is the value of a quantity at a given date, such as a stock price of \$100. A change subtracts the earlier level from the later one. A return measures an investment's gain or loss relative to its starting value. Ignoring dividends, a price rising from \$100 to \$105 has a simple return of $(105 - 100)/100 = 0.05$, or 5 percent.

A logarithm rewrites positive numbers on a scale where multiplication becomes addition. A log return is the difference between the natural logarithms of consecutive prices:

$$r_t = \log P_t - \log P_{t-1} = \log(P_t/P_{t-1}).$$

Here P_t is today's price, P_{t-1} is the previous price, and r_t is the price log return. For the same price move, $\log(105/100) \approx 0.04879$, or about 4.88 percent when multiplied by 100. For small changes, the log return and simple return are close. Log returns add across consecutive periods because the intermediate log prices cancel.

Demeaning subtracts an average from each observation. If the available-data average return is 1 percent and today's return is 3 percent, the demeaned return is 2 percentage points. A percentage point is the difference between two percentages. For real-time calculations, estimate that average using data already available. Demeaning centers a series; it does not by itself remove a trend or establish a repeating cycle.

A trend is a sustained movement in a series' level. Seasonality is a pattern tied to a regular calendar interval, such as higher lunch sales every Friday. A cycle is a broader rise and fall without a necessarily fixed calendar length. Noise is irregular movement not explained by the model. Drawing a wavy line does not establish that a predictable cycle exists.

Nominal values are measured in current money. Real values adjust for changes in the price level. Deflation divides a nominal amount by a suitable price index, a measure of prices relative to a reference period. With a price index normalized to 1 in the reference period, nominal revenue of \$110 and an index of 1.10 give real revenue of \$100 in reference-period dollars. The adjustment separates revenue growth due to prices from other changes.

10.2 Random Walks and Stable Statistical Behavior

A random walk adds an unpredictable change, or shock, to the previous level:

$$y_{t+1} = y_t + u_{t+1}.$$

If the next shock has conditional mean zero given today's information, the expected next level is today's level. Its point forecast is the latest-value baseline. The model adds a statement about how changes arise; the baseline itself is simply a forecasting rule.

Variance measures average squared distance from a mean. It describes spread. Covariance measures whether two quantities tend to be above or below their respective means together. Correlation rescales covariance to a number between -1 and 1, when both variances are positive. Positive correlation describes movement in the same direction; negative correlation describes movement in opposite directions. It does not establish causation.

Autocorrelation is correlation between a series and its own lag. A positive lag-one autocorrelation means high values tend to be paired with high next-period values and low values with low next-period values. This describes a pattern across observations, not a guarantee about each day.

Weak stationarity means a series has a time-invariant mean and finite variance, and its covariance with a lagged value depends on the distance between the dates rather than which dates we select. Imagine comparing weeks in January and June: the model assumes the same underlying average, spread, and lag relationship in both periods. Individual observed values still vary. A steadily rising mean or a permanent jump in spread violates this assumption.

10.3 Differencing and the ARIMA Name

Differencing subtracts the previous observation from the current one:

$$\Delta y_t = y_t - y_{t-1}.$$

Levels of 100, 110, and 125 become changes of 10 and 15. Differencing can help remove some forms of nonstationarity, but it does not guarantee a stable model for any given series.

ARIMA names a family of time-series models. AR means autoregressive: use lagged values. I means integrated: model the series after a specified number of differences. MA means moving average of current and lagged shocks in the model. These shocks are the unpredictable parts after accounting for the model's available information.

In ARIMA(p, d, q), p counts autoregressive lags, d counts differences, and q counts lagged shocks. The MA part has a different meaning from the moving average of observed sales introduced earlier. An MA(1) error model can be written as

$$z_t = \mu + u_t + \theta u_{t-1}.$$

Here z_t is the modeled series, possibly after differencing; μ is its constant mean; u_t is its new mean-zero shock; and θ controls how the previous shock carries into the current value. Assume the shocks are uncorrelated across dates and have a common finite variance. This expression combines shocks, rather than averaging the last two observed sales figures. Estimating these models is beyond the first hand-calculated forecasting task.

10.4 Percentage Errors

A percentage error divides the forecast error by the actual value and expresses the result as a percentage. The mean absolute percentage error, or MAPE, averages the absolute percentage misses:

$$MAPE = \frac{100}{N} \sum_{i=1}^{N} \left| \frac{e_i}{y_i} \right| \%.$$

A forecast of 90 when actual sales are 100 misses by 10 percent. A forecast of 9 when actual sales are 10 also misses by 10 percent. The calculation is undefined for a zero actual value and can become very large for values near zero. This makes MAPE unsuitable for many return series. State the unit and scoring rule whenever comparing forecast accuracy.

11 What You Should Be Able to Explain

You should be able to tell a manager what you are predicting, when the prediction is made, which information it uses, and what decision it supports. You should be able to arrange data by entity and date, calculate a latest-value forecast and an average forecast, and compare saved predictions with actual outcomes.

Before Lecture 2, explain why a model that fits past observations closely still needs a test on later observations. You do not need a complicated model to answer: a forecast must work with information available before the answer is known.